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EXPT.NO - 2

TITLE - QUICK SORT

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In [1]: 1 import random
2 import time
3 import matplotlib.pyplot as plt
4
5 def quick_sort(arr):
6     if len(arr) <= 1:
7         return arr
8     pivot = arr[len(arr)// 2]
9     left = [x for x in arr if x < pivot]
10    middle = [x for x in arr if x == pivot]
11    right = [x for x in arr if x > pivot]
12    return quick_sort(left) + middle + quick_sort(right)
13
14 # Input:
15 arr = list(map(int, input("ENTER THE LIST OF ELEMENTS SEPARATED BY SPACES: ").split()))
16 print("INPUT ARRAY: ",arr)
17
18 sorted_arr = quick_sort(arr)
19 print("SORTED ARRAY: ",sorted_arr)
20
21 # MEASURE TIME TAKEN TO SORT
22 start_time = time.time()
23 sorted_arr = quick_sort(arr)
24 end_time = time.time()
25 print("TIME TAKEN TO SORT: ",end_time - start_time, "SECONDS")
26 print("=====")
27
28 # FOR n>5000
29 n_values = [5000, 6000, 7000, 8000, 10000]
30 time_values = []
31
32 for n in n_values:
33     arr = [random.randint(1,9) for _ in range(n)] # GENERATE RANDOM ARRAY
34
35     start_time = time.time()
36     sorted_arr = quick_sort(arr)
37     end_time = time.time()
38
39     time_taken = end_time - start_time
40     print("TIME TAKEN TO SORT", n, "ELEMENTS:", time_taken, "SECONDS")
41     time_values.append(time_taken)
42 print("=====")
43
44 # plotting the graph
45 plt.plot(n_values, time_values, color = 'red',marker = '*', markersize = 10, linewidth = 1, 1
46 plt.xlabel('NUMBERS OF ELEMENTS (n)')
47 plt.ylabel('TIME TAKEN (SECONDS)')
48 plt.title('QUICK SORT TIME COMPLEXITY ANALYSIS')
49 plt.grid(True)
50 plt.show()
```

ENTER THE LIST OF ELEMENTS SEPARATED BY SPACES: 3 67 7 54 457 4 74 8

INPUT ARRAY: [3, 67, 7, 54, 457, 4, 74, 8]

SORTED ARRAY: [3, 4, 7, 8, 54, 67, 74, 457]

TIME TAKEN TO SORT: 0.0 SECONDS

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TIME TAKEN TO SORT 5000 ELEMENTS: 0.002002716064453125 SECONDS

TIME TAKEN TO SORT 6000 ELEMENTS: 0.002020120620727539 SECONDS

TIME TAKEN TO SORT 7000 ELEMENTS: 0.0020775794982910156 SECONDS

TIME TAKEN TO SORT 8000 ELEMENTS: 0.0020606517791748047 SECONDS

TIME TAKEN TO SORT 10000 ELEMENTS: 0.003100872039794922 SECONDS

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SELECTION SORT TIME COMPLEXITY ANALYSIS

